CYLINDER LINERS

block dowels are in place (fig. 7). Four guide studs made locally (½"-13 x 6") with screwdriver slot in end, such as shown in figure 7 will be found essential when locating the plates on the cylinder block.

2. Apply a light even coating of non-hardening cement to each side of gasket. Position new gasket at end of cylinder block, being careful that gasket is aligned with mating holes.

3. Carefully position end plate over temporary guide studs and with blower drive gear opening in rear plate on blower side of the block.

4. Temporarily install bearing (fig. 7) through the smaller of the two cam and balancer shaft holes in end plate to align plate with block. Start six attaching bolts into place and draw up only finger tight. **BE SURE REAR PLATE** is installed with blower drive support bore on blower side of block, also that **SMALL CAM OR BALANCE BEARING HOLE** in front and rear plates are both on same side.

5. The six cap screws may now be tightened and bearing removed. Temporary guide studs can be left in place until flywheel housing has been installed.

CYLINDER LINERS

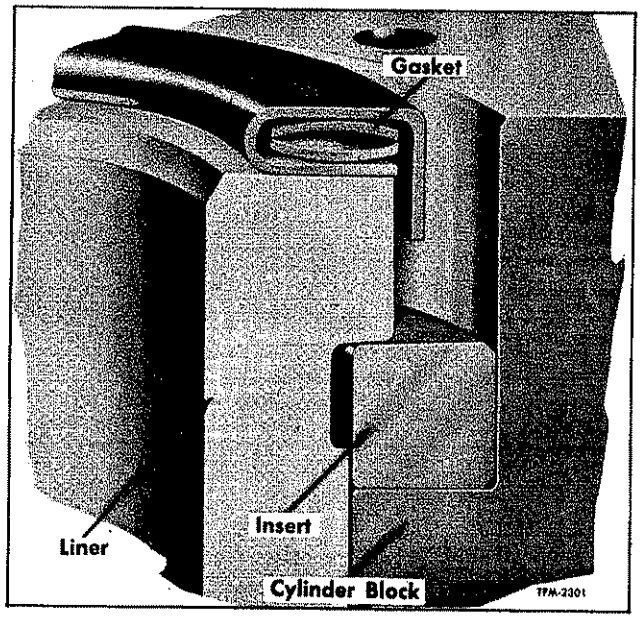


Figure 8—Cylinder Liner, Insert, and Gasket Mounting in Block

GENERAL

Construction of cylinder block, cylinder head, and liner provides that top of cylinder liner flange be below the top of cylinder block and compression seal is accomplished by use of individual gaskets at each cylinder (fig. 8), thus cylinder head rests directly on cylinder block and is called metal-to-metal contact. Water and oil passages between cylinder block and cylinder head are individually sealed with special grommet type rubber gaskets which are installed in cylinder block counterbores.

The replaceable liner, made of hardened alloy cast iron, in each cylinder is accurately honed to a very smooth finish. Even temperature and minimum

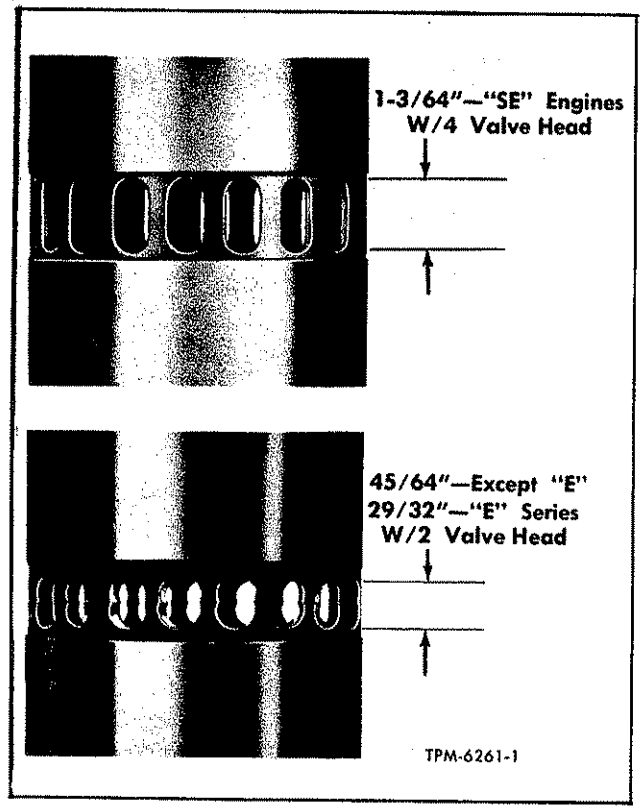


Figure 9—Cylinder Liner

distortion are insured by cooling each liner over its entire length, except at the ports, which are cooled by the scavenging air. To permit introduction of fresh air into the cylinder, twenty ports, with a figure 8 appearance, are drilled into the circumference of each cylinder liner. Refer to figure 9.

NOTE: Cylinder liners with three sizes of ports are used. Liner used in "E" series engines with two valve heads measure .900" (approx. $^{29}/_{32}$") high, or